

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	01 July 2026
Team ID	
Project Name	Nutrition Assistant using MERN stack
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion	Connect 345,408 row Excel dataset to Tableau Verify data types and geographic roles
FR-2	Visualization Generation	Create State-wise Choropleth Map Create Year-wise trend analysis chart Create Crop-wise bar charts
FR-3	Interactivity	Implement cross-filtering across all 6 filters (State, Crop, Season, etc.) Configure dynamic KPI cards
FR-4	Web Integration	Embed Tableau Dashboard via Iframe Develop responsive HTML/CSS/Bootstrap frontend

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard must be responsive and intuitively usable on desktop and tablet devices without training.

NFR-2	Security	Web application must be deployed over HTTPS. No user PII is collected.
NFR-3	Reliability	Platform should achieve 99% uptime relying on Vercel and Tableau Public infrastructure.
NFR-4	Performance	Dashboard filters should render updates across 345,408 rows within 3-5 seconds.
NFR-5	Availability	Accessible globally 24/7 without login requirements.
NFR-6	Scalability	Architecture must support concurrent public users via Tableau CDN.